**Q1.** (a) The table gives some of the properties of three substances: water, paraffin wax and naphthalene.

| substance | melting point in °C | density (mass of 1 $cm^3$ in g) |
|---|---|---|
| water | 0 | 1.0 |
| paraffin wax | 60 | 0.8 |
| naphthalene | 80 | 1.2 |

**Use the information in the table to help you answer parts (i) and (ii) below.**

(i) Arif placed a ball of naphthalene in a beaker containing water kept at 70°C.
On diagram 1, draw the **shape** and **position** of the naphthalene ten minutes later.

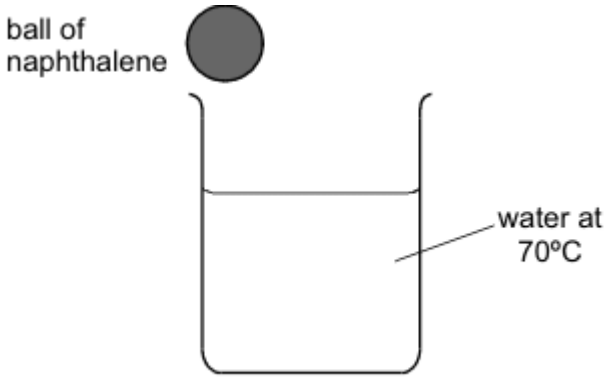



**diagram 1**

2 marks

(ii) Arif repeated the experiment with a similar sized ball of paraffin wax.
On diagram 2, draw the **shape** and **position** of the paraffin wax ten minutes later.

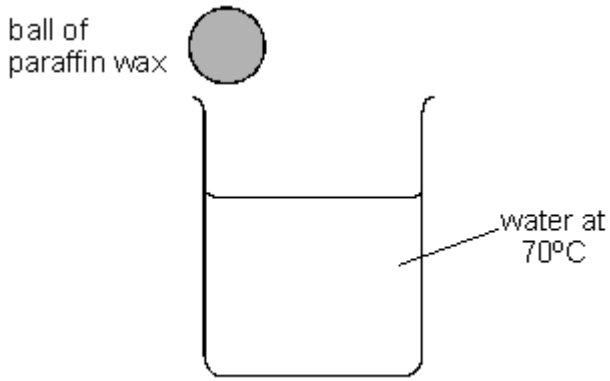



**diagram 2**

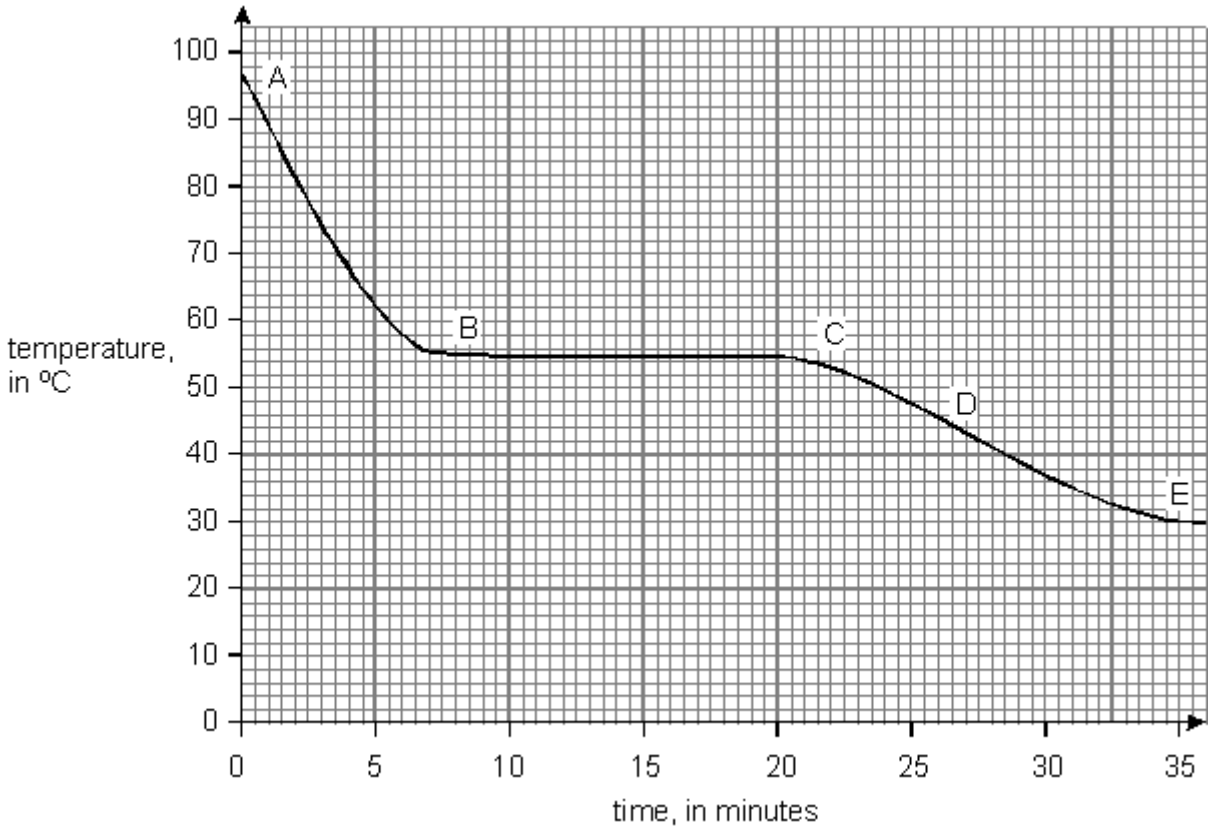
2 marks

(b) Arif then melted some crystals of stearic acid in a beaker. He placed a temperature probe in the stearic acid and left the stearic acid to cool.
As the stearic acid cooled down its state changed.
What was the change in the state of the stearic acid as it cooled?

From …… to ……

1 mark

(c) Arif plotted the following graph.




Which letter on the graph shows when the stearic acid **started** to change state?

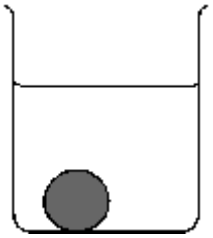
……

1 mark
Maximum 6 marks

Mark schemes

## Q1.

(a) (i)

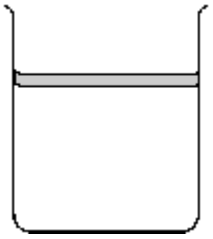


*one mark is for a ball of solid naphthalene*

*one mark is for the naphthalene at the bottom of the beaker*

**2 (L5)**

(ii)



*one mark is for a layer of melted wax*

*one mark is for showing the wax floating on top of the water*

**2 (L6)**

(b) *from* liquid *to* solid

*answers must be in the correct order*

**1 (L5)**

(c) B

**1 (L6)**

**[6]**